AI-Generated Content Detection Pattern

Detecting Human vs. ChatGPT-Written Answers Using the HC3 Dataset

Project Report

Mini Project — Data Analysis with Python

Abstract

Large language models such as ChatGPT can now produce answers that read as fluently as human writing, making it important to understand what, if anything, still separates the two. This project uses the HC3 (Human ChatGPT Comparison Corpus) dataset to build a clean, balanced collection of human- and ChatGPT-written answers to the same questions, and explores simple, interpretable text statistics — word count, sentence length, first-person usage and question marks — that differ measurably between the two sources. The result is a 52,346-answer dataset, balanced 50/50 between classes, along with exploratory visualizations that highlight clear, consistent stylistic differences and lay the groundwork for a future classification model.

Introduction

The rise of conversational AI has made it increasingly difficult to tell whether a piece of text was written by a human or generated by a model like ChatGPT. This has implications for academic integrity, misinformation, and content moderation. Rather than jumping straight to a black-box classifier, this project first asks a simpler question: are there measurable, human-interpretable patterns — length, structure, tone — that distinguish human answers from ChatGPT answers? Answering this well requires a clean, well-structured dataset, which is the main deliverable of this project, alongside the exploratory analysis that surfaces those patterns.

Dataset Description

The dataset used is **HC3** (Hello-SimpleAI/HC3) from Hugging Face, specifically the **"all"** subset. Each row pairs a question with one or more human answers and one or more ChatGPT answers, drawn from five sources: Reddit ELI5, finance, medicine, open QA, and Wikipedia CS/AI topics. The raw file contains 24,322 question rows before cleaning. ChatGPT answers in this dataset were generated by GPT-3.5-era models, and the majority of questions originate from the Reddit ELI5 community.

Data Cleaning

- **1. Drop unused column and duplicates**
  Dropped the mostly-empty **index** column and removed 509 exact duplicate rows. **Before:** 24,322 rows → **After:** 23,813 rows.

- **2. Parse and explode answer lists**
  Each row's **human_answers** and **chatgpt_answers** columns held JSON-encoded lists of answers. These were parsed and exploded so every row holds exactly one answer, labeled 0 (Human) or 1 (ChatGPT), then null and blank answers were removed. **Before:** 23,813 question rows → **After:** 83,400 answers (57,019 Human, 26,381 ChatGPT).

- **3. Add text-statistics features**
  Computed **word_count**, **sentence_count**, **avg_sentence_len**, **first_person** count (I / me / my), and **question_marks** count for every answer.

- **4. Fix spacing artifacts and remove giveaway phrases**
  Removed stray whitespace before punctuation and contractions (e.g. "do n't" → "don't"), and stripped the literal phrase "As an AI language model" that some ChatGPT answers contained verbatim.

- **5. Remove very short / duplicate answers**
  Dropped answers under 20 words and duplicate answer text. **Before:** 83,400 answers → **After:** 75,191 answers (49,018 Human, 26,173 ChatGPT).

- **6. Balance the two classes**
  Down-sampled the majority (Human) class to match the minority (ChatGPT) class size, using **random_state=42** for reproducibility. **Before:** 49,018 Human / 26,173 ChatGPT → **After:** 26,173 Human / 26,173 ChatGPT, saved as **cleaned_dataset.csv** (52,346 rows total).

Exploratory Data Analysis

With one answer per row, we examined how word count and writing-style features differ between Human and ChatGPT answers. Human answers vary far more in length (std. dev. ≈ 164 words, driven by a long tail of very short and very long Reddit replies) while ChatGPT answers cluster tightly around a higher average length. ChatGPT also tends to write in longer, more uniform sentences, while human answers use first-person language and embedded questions much more frequently — both signs of a more conversational, personal writing style.

Word count by label: Human — mean 134.0, median 83, std 163.7. ChatGPT — mean 173.1, median 174, std 61.6.

Writing-style feature means by label (Human vs ChatGPT): sentence count 7.75 vs 9.15; average sentence length 16.15 vs 19.11 words; first-person usage 0.95 vs 0.21 per answer; question marks 0.22 vs 0.07 per answer.

Visualizations

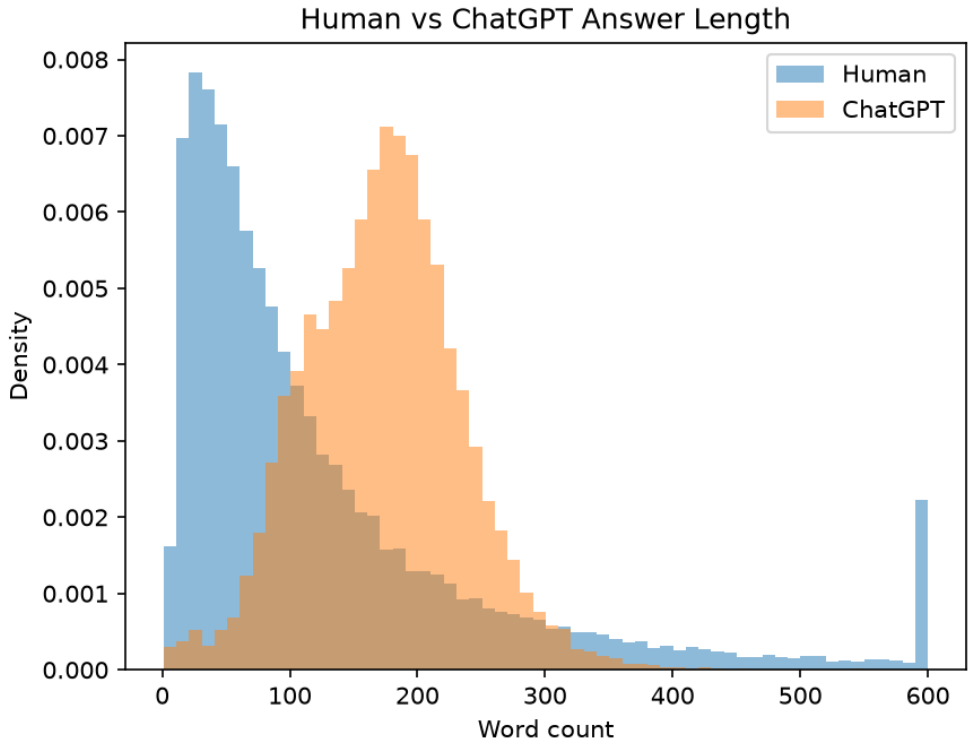


Figure 1. Distribution of answer length (word count, clipped at 600) by label.

ChatGPT answers form a tight, bell-shaped distribution centered around 150–200 words, while human answers are heavily skewed toward short replies with a long tail — reflecting the free-form, variable-length nature of human writing versus the more templated length of ChatGPT responses.

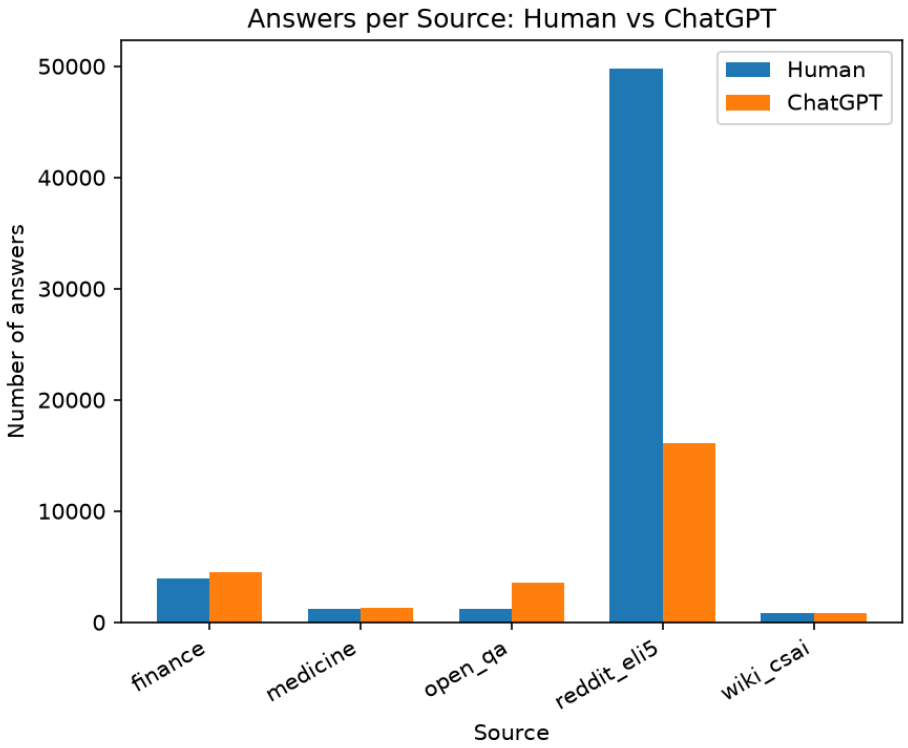


Figure 2. Number of answers per source, split by Human vs ChatGPT.

Reddit ELI5 dominates the dataset and contributes far more human answers than ChatGPT answers per question (each ELI5 question has up to 3 human replies but 1 ChatGPT reply), while finance, medicine, open QA and wiki_csai are comparatively small and more evenly split between the two labels.



Figure 3. Average writing-style features by label.

ChatGPT consistently produces longer sentences and slightly more of them, while human answers show far higher first-person usage and more embedded questions — a quantitative signature of the conversational, first-hand tone typical of human replies.

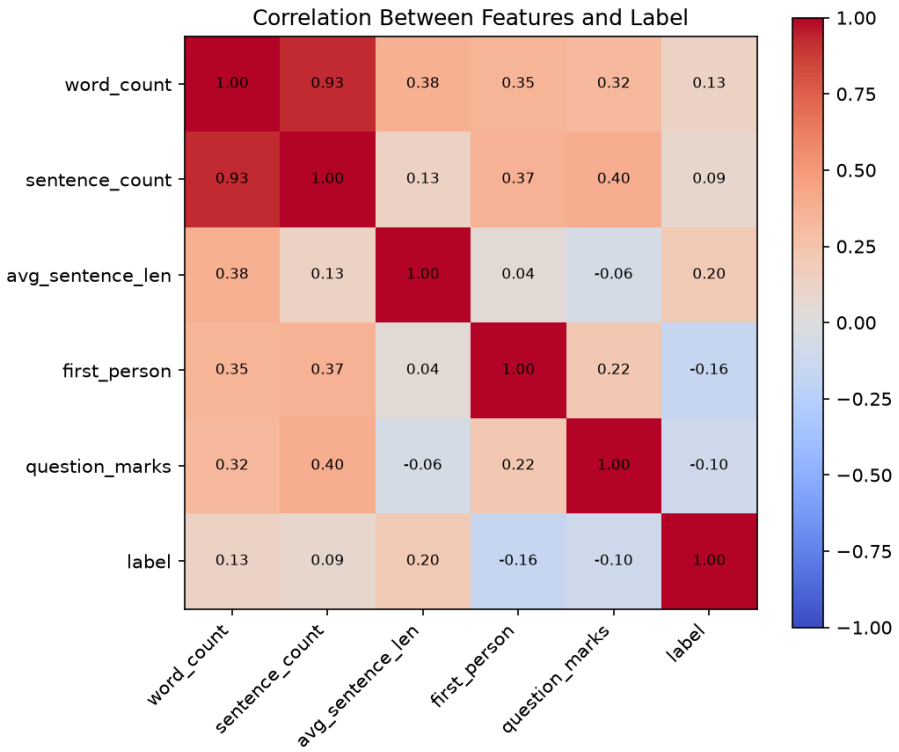


Figure 4. Correlation heatmap between word count, writing-style features, and label.

Word count and sentence count are strongly correlated (0.93), as expected. The label correlates most with first_person (-0.16) and avg_sentence_len (0.20), confirming these as the most discriminative individual features, though no single feature is strongly correlated with the label on its own — suggesting a combination of features would be needed for reliable classification.

Key Findings

- ChatGPT answers are longer (median 174 words vs 83 for human) and much more consistent in length.
- Humans use first-person words (I, me, my) about 4.5 times more often (0.95 vs 0.21 per answer).
- Humans ask more questions inside answers (0.22 vs 0.07 question marks per answer).
- ChatGPT writes longer sentences (19.1 vs 16.1 words per sentence).
- Final dataset: 52,346 answers, balanced 50/50, saved as cleaned_dataset.csv.

Limitations

- The ChatGPT answers in HC3 come from a GPT-3.5-era model; newer models may write with different, less distinguishable patterns, so these findings may not generalize to current-generation LLMs.
- Reddit ELI5 dominates the source distribution, so overall statistics are weighted heavily toward conversational, informal Reddit-style writing rather than the other four domains.
- Any unique-word or vocabulary-diversity style measure is sensitive to text length, so such measures would need length normalization before being compared fairly across Human and ChatGPT answers.

Future Work

The natural next step is to build a classification model on top of this cleaned, balanced dataset. A reasonable baseline would be a **TF-IDF + Logistic Regression** pipeline on the answer text, which should perform well given the clear stylistic differences surfaced in the EDA. To get a realistic estimate of generalization, the train/test split should be done **by question** rather than by answer, so that human and ChatGPT answers to the same question never leak across the split. Future iterations could also test performance separately per source, and evaluate against answers from newer-generation models.

Conclusion

This project transformed a raw, nested question-answer corpus into a clean, balanced, answer-level dataset suitable for analysis and modeling. Even without a trained classifier, simple text statistics — answer length, sentence length, first-person usage, and question marks — reveal clear and consistent differences between human and ChatGPT writing styles. These findings validate that AI-generated content detection is a tractable problem and provide a solid, well-understood foundation for the modeling work proposed above.